

AI 200 Linear Algebra, Eigenvalue Decomposition, Assignment *

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1. Let $\mathbf{A} \in \mathbb{R}^{N \times N}$ be a real-valued square matrix. Suppose it has N distinct eigenvalues $\lambda_0, \dots, \lambda_{N-1} \in \mathbb{C}$ and N associated eigenvectors $\mathbf{x}_0, \dots, \mathbf{x}_{N-1} \in \mathbb{C}^{N \times 1}$.

Prove with math induction that all these eigenvectors are linearly independent.

2. For real-valued square matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$, denote

$$\mathbf{Q} = [\mathbf{q}_0 \quad \mathbf{q}_1 \quad \cdots \quad \mathbf{q}_{N-1}]$$

where each column $\mathbf{q}_n \in \mathbb{R}^{N \times 1}$ is an eigenvector associated with eigenvalue λ_n .

Denote

$$\mathbf{\Lambda} = \begin{bmatrix} \lambda_0 & & \\ & \ddots & \\ & & \lambda_{N-1} \end{bmatrix}.$$

Prove the following statements:

(a)

$$\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{-1}.$$

(b) Define

$$\mathbf{P} = [\mathbf{p}_0 \quad \cdots \quad \mathbf{p}_{N-1}] = \mathbf{Q}^{-1, \top}.$$

Then

$$\mathbf{A} = \sum_{n=0}^{N-1} \lambda_n \mathbf{q}_n \mathbf{p}_n^{\top}.$$

3. Let a real-valued square matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$ be with the following eigenvalue decomposition:

$$\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{-1}.$$

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Define

$$\mathbf{P} = \begin{bmatrix} \mathbf{p}_0 & \cdots & \mathbf{p}_{N-1} \end{bmatrix} = \mathbf{Q}^{-1, \top}.$$

Prove the following statements:

- (a) $\mathbf{A}^\top$ has the following eigenvalue decomposition:

$$\mathbf{A}^\top = \mathbf{P} \mathbf{\Lambda} \mathbf{P}^{-1}.$$

- (b) We have

$$\mathbf{A}^\top \mathbf{p}_n = \lambda_n \mathbf{p}_n, \quad \forall n \in \{0, \dots, N-1\},$$

4. Let

$$\mathbf{A} = \begin{bmatrix} 2 & -1 & 1 & 5 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 4 & 7 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

- (a) Manually compute eigenvalues and right and left eigenvectors of this eigenvalue equation. (In case there are eigenvectors associated with the same eigenvalue, ensure these eigenvectors are orthogonal to each other.)
- (b) Manually do eigendecomposition of $\mathbf{A}$.
5. (a) Given $\mathbf{A}$ in Problem 4, use the eigendecomposition found in that problem to compute
- $$4\mathbf{A}^3 - 3\mathbf{A}^2 + 5\mathbf{A} + 1.$$
- (b) Verify your solution with `np.linalg.matrix_power`.
6. Given $\mathbf{A}$ in Problem 4, use the eigendecomposition found in that problem to solve the following differential equation:

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x},$$

with the initial value $\mathbf{x}(0)$.

7. Consider the following Markov transitional matrix from states in period t to states in period $t+1$:

$$\mathbf{P} = \begin{bmatrix} 0.2 & 0.3 & 0.5 \\ 0.4 & 0.5 & 0.1 \\ 0.6 & 0.2 & 0.2 \end{bmatrix}.$$

That is, $p_{ij} = \mathbb{P}(X_{t+1} = j | X_t = i)$, for all $t \geq 0$.

Suppose at time 0, the probability of being at state i is $p_{0,i}$. Denote

$$\mathbf{p}_0 = \begin{bmatrix} p_{0,0} \\ p_{0,1} \\ p_{0,2} \end{bmatrix}.$$

- (a) Prove by math induction that the probability distribution vector on different states in period t is given by

$$\mathbf{p}_t = \mathbf{P}^t \mathbf{p}_0.$$

- (b) Do eigendecomposition of matrix $\mathbf{P}$ (Hint: One eigenvalue is 1).
 (c) Use the eigendecomposition of matrix $\mathbf{P}$ to derive a closed-form of $\mathbf{p}_t$.
 (d) Compute

$$\lim_{t \rightarrow \infty} \mathbf{p}_t.$$

Does your solution depend on the initial condition $\mathbf{p}_0$? Please interpret this result from the geometric perspective of the eigendecomposition.

8. This is a coding task.

Use NumPy to write a function called `power_dom_eigen` that uses the power method to compute the dominant eigenvalue of a real-valued square matrix:

- (a) Input: a real-valued square matrix with shape (N,N) . You are guaranteed that the input matrix has all real and distinct eigenvalues.
 (b) Output: three NumPy objects.
 i. Eigenvalue with dimension 0.
 ii. Right eigenvector with shape $(N,)$.
 iii. Left eigenvector with shape $(N,)$.

9. This is a coding task.

Use NumPy to write a function called `power_top_eigen` that uses the power method and the deflated matrix formula to compute all eigenvectors.

- (a) Input:
 i. A real-valued square matrix with shape (N,N)
 ii. The number top eigenvalues that shall be computed (in terms of their absolute values), with the argument name K .
 iii. You are guaranteed that the input matrix has all real and distinct eigenvalues.
 (b) Output: three NumPy objects.
 i. Top eigenvalues with shape $(K,)$.
 ii. Associated right eigenvectors with shape (K,N) .
 iii. Associated left eigenvector with shape (K,N) .

- (c) In the body of this function, you can call function `power_dom_eigen` that you defined in Problem 8.
- 10. (a) Use the function you defined in Problem 9 to do eigendecomposition of the matrix defined in Problem 7.
- (b) Verify your solution with `np.linalg.eig`.

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